

LogiSense

Intelligent Logistics, Data Warehousing & Predictive Delivery Analytics

Project README / Documentation

LogiSense is a logistics intelligence platform combining a data warehouse, OLAP analytics, machine learning, advanced data mining, and a web application into one end-to-end decision-support system.

The pipeline is: **Raw Data** → **ETL** → **Star Schema DWH** → **OLAP/BI** → **ML & Data Mining** → **Web Application** → **Operational and Customer Decisions**.

1. Project Overview

LogiSense analyzes e-commerce delivery and shipping data to answer practical logistics questions: which shipments are likely to be late, how long deliveries are expected to take, which carriers and warehouses perform best, and how outcomes vary by geography, product, shipping method, weather, priority and customer segment.

2. Objectives

- Build a useful real-world logistics analytics platform.
- Implement a star-schema data warehouse and ETL pipeline.
- Support OLAP slice, dice, roll-up and drill-down.
- Predict late-delivery risk before the outcome is known.
- Forecast delivery duration with regression.
- Apply clustering, association-rule mining and anomaly detection.
- Provide an enterprise-style dashboard and role-based customer experience.

3. Dataset

The project uses a Kaggle e-commerce delivery/shipping dataset containing **50,000 rows and 30 columns**. It contains order, customer, warehouse, product, shipping, carrier, distance, promised/actual delivery, cost, package, payment, tracking, weather, rating, return, processing and priority information.

ML leakage prevention: `actual_delivery_days`, `delivery_attempts`, `delivery_delay_days`, `delivery_status`, `late_delivery` and `tracking_status` are not used as prediction inputs because they expose information unavailable at prediction time.

4. Data Warehouse — Star Schema

The central fact table has the grain **one row per delivery order**. Dimension tables provide descriptive context for analytical queries.

Table	Role	Purpose
fact_delivery	Fact	Delivery-order measures and foreign keys
dim_date	Dimension	Calendar attributes
dim_customer	Dimension	Customer and segment attributes
dim_product	Dimension	Product/category attributes
dim_shipping	Dimension	Shipping method attributes
dim_carrier	Dimension	Carrier information
dim_warehouse	Dimension	Warehouse/location information
dim_weather	Dimension	Weather context

5. Notebooks

- 01_Data_Audit_EDA_Business_Intelligence.ipynb — data audit, EDA and BI.
- 02_Data_Warehouse_Star_Schema_ETL.ipynb — ETL and warehouse construction.
- 03_OLAP_Explorer_Slice_Dice_Rollup_Drilldown.ipynb — OLAP operations and queries.
- 04_ML_Classification_Regression.ipynb — classification and regression.
- 05_Advanced_Data_Mining_Clustering_Association_Anomaly.ipynb — K-Means, Apriori and Isolation Forest.
- 06_Executive_Analytics_Decision_Simulation.ipynb — executive analytics and decision simulation.

6. OLAP Explorer

Dimensions include Country, Customer Segment, Warehouse, Carrier, Product Category, Shipping Method, Weather, Priority, Year, Quarter and Month. Measures include Orders, Order Value, Shipping Cost, Average Delivery Days, Average Delay Days, Average Rating, Late Rate and Return Rate.

Slice: select one dimension value. **Dice:** filter multiple dimensions. **Roll-up:** aggregate from detail to summary. **Drill-down:** move from summary to finer detail.

7. Machine Learning

Model	Task	Reported performance
Late Delivery Risk	Classification	Accuracy 65.9%; Precision 70.6%; Recall 67.4%; F1 69.0%; ROC-AUC 71.9%; threshold 0.35
Delivery Time	Regression	MAE 1.17 days; RMSE 1.63 days; R² 0.876

Saved artifacts: `models/late_delivery_risk_model.joblib`, `models/delivery_time_model.joblib` and `models/model_metadata.json`.

Approved pre-outcome features include customer segment/city/country, warehouse, product category, shipping method, carrier, package size, payment method, product weight, order value, distance, promised delivery days, shipping cost and derived order-date features (year, month, day, day-of-week and weekend flag).

8. Advanced Data Mining

- **K-Means clustering** groups similar logistics/order behavior.
- **Apriori association mining** discovers recurring relationships among categorical attributes and outcomes.
- **Isolation Forest** identifies unusual or anomalous delivery patterns.

9. Web Application

The frontend is a React + TypeScript + Vite application with an enterprise dashboard design. Flask provides backend/API and ML integration.

Major areas include Dashboard, OLAP Explorer, Delivery Analytics, Carrier Analytics, Risk Prediction, Delivery Forecast, Orders Explorer, Data Quality, Model Monitoring, Admin Portal, Customer Portal and Settings.

10. Role-Based Access

LogiSense provides a unified login experience with role-specific access. Administrators use the operations/admin experience, while customers use the customer portal for delivery-focused interactions.

11. Project Structure

```
LogiSense/
├── app/           Flask application and ML/API integration
├── frontend/      React + TypeScript + Vite frontend
├── data/raw/      Raw source data
├── data/processed/ Star-schema CSV tables
├── models/        Saved ML models and metadata
├── notebooks/     DWM, OLAP, ML and mining notebooks
├── backend/       Backend/API assets
├── database/      Database assets
└── etl/           ETL assets
```

■■■ ml/	ML assets
■■■ tests/	Tests
■■■ docs/	Documentation

12. End-to-End Architecture

Raw E-commerce Data → **Audit/EDA** → **ETL** → **Star Schema DWH** → **OLAP/BI** → **ML/Data Mining** → **Flask APIs** → **React Application** → **Admin & Customer Decisions**

13. Running the Project

Backend

```
source .venv/Scripts/activate
python -m app.app
```

Frontend

```
cd frontend
npm install
npm run dev
```

Run the backend and frontend in separate terminals using the repository's existing configuration.

14. Business Value

LogiSense can identify potentially late deliveries, forecast delivery duration, compare carriers and warehouses, analyze operational trends, detect unusual patterns, support evidence-based logistics decisions, and provide customers with delivery-focused visibility.

15. DWM Coverage

The project demonstrates ETL, dimensional modeling, star schema, fact/dimension tables, OLAP, slice, dice, roll-up, drill-down, business intelligence, classification, regression, clustering, association-rule mining, anomaly detection, model evaluation, leakage prevention and analytical-model integration.

16. Key Technical Notes

- Fact-table grain: one row per delivery order.
- ML uses pre-outcome features only.
- Classification threshold: 0.35.
- Analytics, warehouse data, models, backend and frontend are separated into project layers.